PYCLIMAEXPLORER: DECODING OUR PLANET'S CLIMATE

A Rapid-Prototype Interactive Visualizer for Climate Model Data

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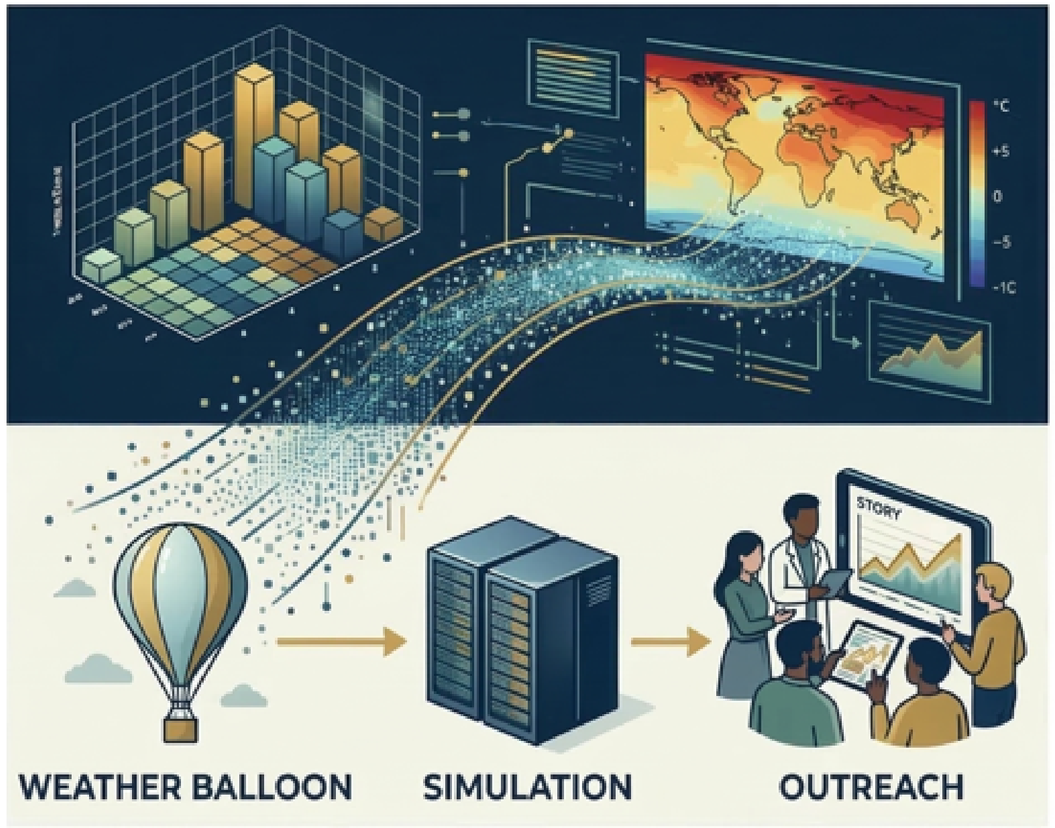
BRIDGING THE GAP IN CLIMATE DATA

The Challenge: Visualizing simulation output is crucial for scientific understanding and public outreach.
However, massive, multi-dimensional climate datasets are often inaccessible to non-experts.

Our Vision: To build a tool that turns raw numbers into a compelling story about our changing planet.

Target Audience: Researchers needing quick looks at data, and the general public interested in climate trends.



PROPOSED SOLUTION

PyClimaExplorer is an interactive web-based dashboard designed to simplify the exploration of complex climate datasets stored in NetCDF format.

Key Idea: Transform multi-dimensional climate data into intuitive and interactive visual insights.

Core Capabilities

- **Interactive Data Exploration:** Users can dynamically select climate variables such as Surface Temperature, Precipitation, or Wind Speed.
- **Flexible Time Selection:** Filter and visualize data across different time ranges to observe climate trends
- **Spatial Visualization (Map View):** Display climate variables on a global map, showing geographical distribution for a selected time slice.
- **Temporal Visualization (Time-Series View):** Generate time-series graphs to analyze how a climate variable changes over time at a specific location.

THE PYTHON DATA ECOSYSTEM

Technology stack

- **Data handling**
  - **Xarray** - for multi-dimensional climate datasets(NetCDF)
  - **Pandas**-efficicent time -series data manipulation
  - **NumPy**-high-performance numerical computations
- **Web Framework**
  - **Streamlit**-rapid prototyping speed and clean UI).
- **Visualization engine**
  - **Plotly** - interactive charts and mas for dynamic climate visualization
  - **Matplotlib**-additional support for scientific plotting
- **Optimization**
  - **Xarray's Lazy Loading**- enables efficient handling of large NetCDF datasets without loading entire files into memory

SYSTEM ARCHITECTURE

Ingestion

Loading nc files with lazi loading for large sets.

Processing

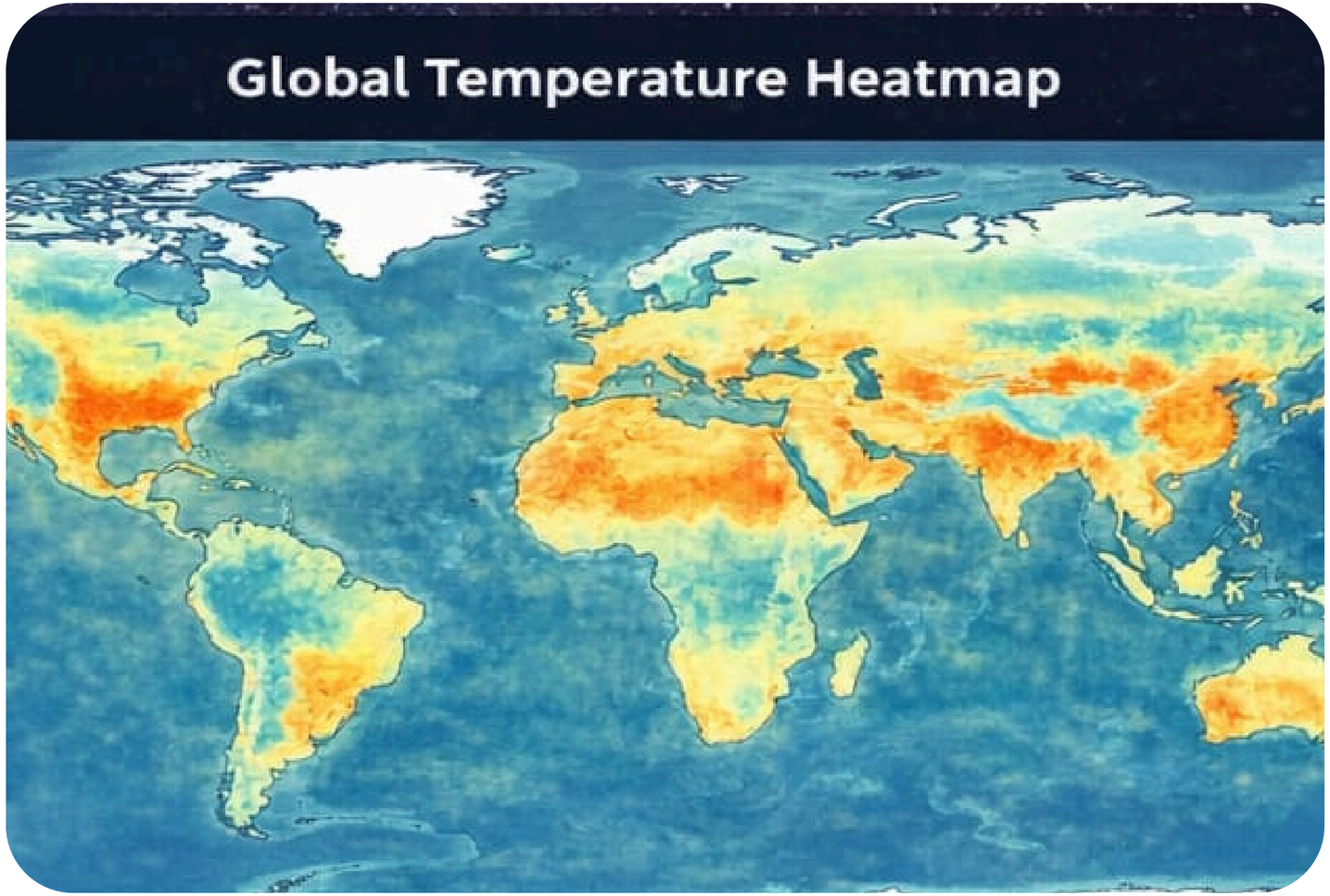
Vectorized operation for coordinate filtering

Analytics

Anomaly detection and localized trend extraction

Dashboard

Streamlit cloud-deployed interactive UI



KEY FEATURES

Spatial Visualization

- Interactive global heatmap of climate variables
- Shows distribution of temperature, precipitation, and wind speed

Temporal Visualization

- Time-series charts for selected locations
- Analyze climate trends over time

Interactive Controls

- Variable selection (Temperature, Precipitation, Wind)
- Time range filtering
- Location selection on map

Proposed Enhancements

- 3D Globe Visualization using PyDeck
- Dataset Comparison Mode (e.g. 1990 vs 2020)
- Story Mode highlighting important climate events

IMPACT

Why PyClimaExplorer Matters?

- **Simplifies Complex Climate Data**
  Transforms large multi-dimensional NetCDF datasets into intuitive visual insights.
- **Enables Rapid Data Exploration**
  Researchers can quickly analyze climate variables through interactive maps and time-series plots.
- **Improves Accessibility**
  Bridges the gap between scientific research and public understanding of climate trends.
- **Future Opportunities**
  - Real-Time Satellite Data Integration for live climate monitoring
  - AI-Driven Anomaly Detection to identify unusual climate patterns
- **Mission**
  To transform raw climate data into clear, interacting visuals stories that help understand our changing planet.